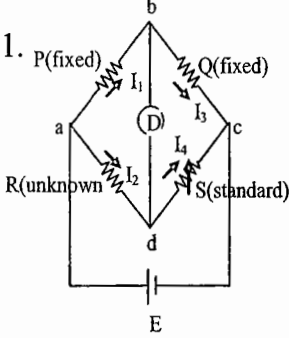
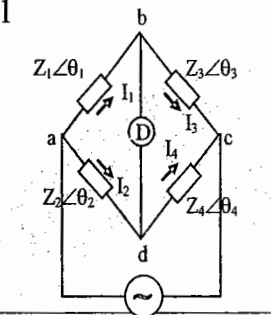


4

Bridge Measurement of R, L & C

There are two types of bridges

1. DC Bridges
2. AC Bridges

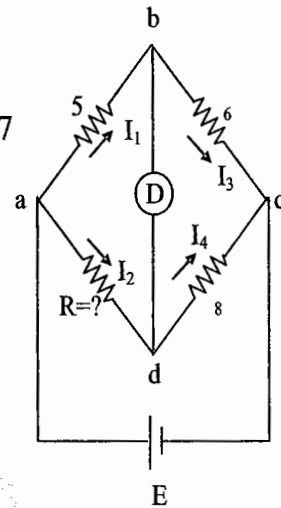
DC Bridge	AC Bridge
1. 	1. 
2. source: Battery	2. AC source, oscillator
3. Detector: DC Galvanometer (0Hz)	3. Vibrational galvanometer (5-20Hz) • Telephone detectors in head phones (200-5kHz) • Tuned amplifier (>5kHz)
4. Balanced PS = QR	4. $Z_1 Z_4 = Z_2 Z_3$ and $\theta_1 + \theta_4 = \theta_2 + \theta_3$
5. Application: only 'R' measurement	5. L, C, f, Q-factor, D-factor, & ϵ_r

Note: Don't connect Galvanometer and source to the same points. It may be unbalance conditions.

- If the below Bridge is balanced then the value of R is
PS = QR

$$5 \times 8 = 6 \times R$$

$$R = \frac{5 \times 8}{6} = 6.67$$



Example 1:

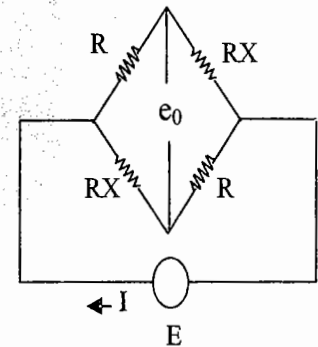
For the circuit given in figure the unbalance voltage e_0 is

(A) $e_0 = \frac{IRX}{2}$

(B) $e_0 = 2IRX$

(C) $e_0 = IR(1+X)$

(D) $e_0 = \frac{IR(X-1)}{2}$



Ans: (D)

Sol: equivalent resistance = $(R + RX) // (R + RX)$

$$= \frac{(R + RX)}{2}$$

$$\therefore I = \frac{E}{(R + RX)/2}$$

$$\Rightarrow E = I \frac{(R + RX)}{2}$$

$$e_0 = E \left[\frac{RX}{R + RX} \right] - E \left[\frac{R}{R + RX} \right]$$

$$= E \left[\frac{RX - R}{R + RX} \right] = \frac{IR(X-1)}{2}$$



When bridge is balanced, adjacent branch voltages are equal.

- To reduce the sliding balance, select uncommon elements as variables always (R_4 & C_4).
- If the selected uncommon elements are present in same arm. Then time taken for bridge balance is much reduced then it is called converges to sliding balance.

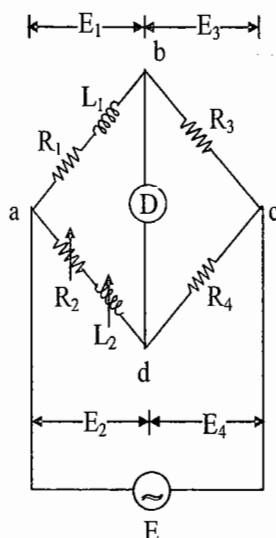
• **Measurement of Inductance :**

- 1) Maxwell – Inductance Bridge
- 2) Maxwell – Inductance –capacitance bridge (medium Q coils $1 < Q < 10$)
- 3) Hays Bridge – (High Q coils $Q > 10$)
- 4) Owen's Bridge – (Incremental Inductance (μ_0))
- 5) Anderson's bridge – ("low Q coils" ($Q < 1$))

(accurate measurement of Inductor from milli Henry to wide range)

⇒ out of above 5 bridges Anderson bridge is most accurate bridge.

Maxwell –Inductance Bridge:



$$(R_1 + j\omega L_1) \times R_4 = (R_2 + j\omega L_2) \times R_3$$

Under balanced conditions

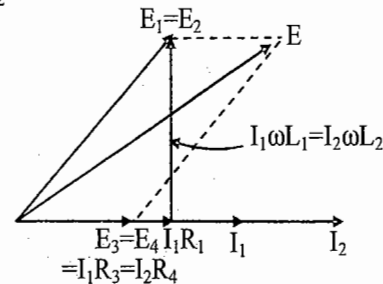
$$R_1 R_4 = R_2 R_3$$

$$R_1 = \frac{R_2 R_3}{R_4}$$

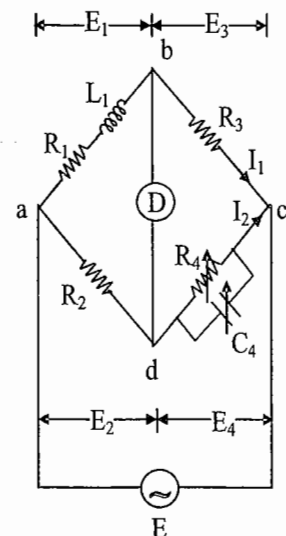
$$L_1 R_4 = R_2 R_3$$

$$L_1 = \frac{L_2 R_3}{R_4}$$

To make Bridge balance select uncommon elements from the above two equations those are " R_2 & L_2 "



Maxwell –Inductance –Capacitance Bridge:



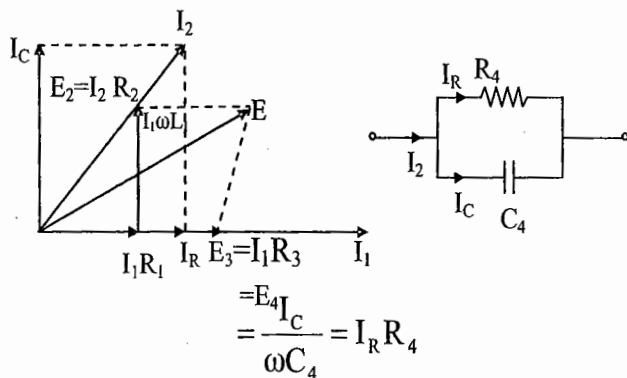
$$R_1 R_4 = R_2 R_3$$

$$R_1 = \frac{R_2 R_3}{R_4}$$

$$L_1 = R_2 R_3 R_4$$



$$Q \text{ factor} = \frac{\omega L_1}{R_1} = \omega R_4 C_4$$



$$|V_4| = I_R R_4 \rightarrow I_R \text{ in phase } |V_4|$$

$$= \frac{I_C}{\omega C_4} \rightarrow I_C \text{ leads } V_4 \text{ by } 90^\circ$$

$$I_2 = I_R + I_C$$

- It's not suitable for high & low 'Q' coils it's used for measurement of a medium Q coils Inductance & resistance from a known value of capacitance

$$Q = \frac{\omega L_1}{R_1} = \frac{X_{L1}}{R_1} \rightarrow \text{high} \approx \infty$$

$$Z_1 = R_1 + j X_1$$

$$\theta_1 = \tan^{-1} \left(\frac{X_{L1}}{R_1} \right)$$

$$\theta_1 = 90^\circ$$

$$\theta_1 + \theta_4 = \theta_2 + \theta_3$$

$$\theta_4 = -90^\circ$$

For high 'Q' coil

$$\tan^{-1}(\omega C_4 R_4) = 90^\circ$$

$$(\omega C_4 R_4) \leq \infty \rightarrow (\text{Not practical})$$

Hence this bridge is not used for High Q factor measurement

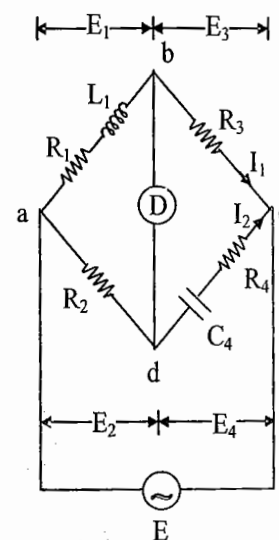
If low Q coil:

$$\theta_1 = \tan^{-1}(0) = 0$$

$$\text{So } \omega C_4 R_4 = 0$$

So also we can't change R_2 & R_3 values to this low values. So bridge is not suitable for low Q_1 factor measurement.

Hays Bridge:



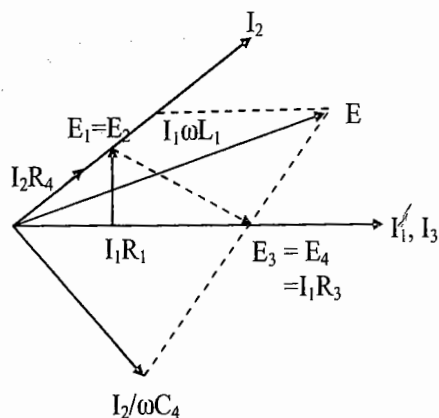
Under bridge balance condition

$$Z_1 Z_4 = Z_2 Z_3$$

$$L_1 = \frac{R_2 R_3 C_4}{(1 + \omega^2 R_4^2 C_4^2)}$$

$$R_1 = \frac{\omega^2 R_2 R_3 R_4 C_4^2}{(1 + \omega^2 R_4^2 C_4^2)}$$

$$Q = \frac{\omega L_1}{R_1} = \frac{1}{\omega R_4 C_4}$$



Phasor diagram

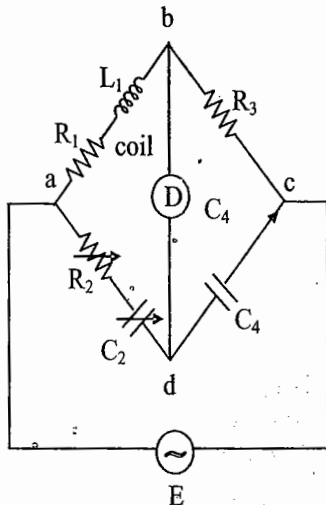
This is the "slowest bridge"

- used only for "high Q-factor measurement"



Owen's bridge:

This is the bridge which is having two capacitances in the bridge.



Under balanced conditions

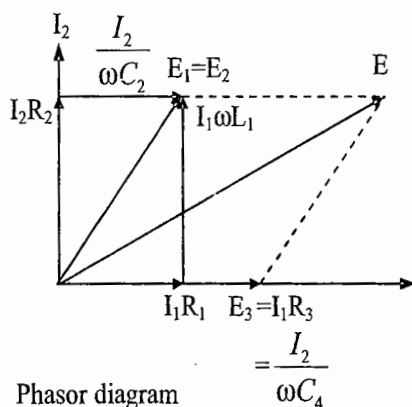
$$Z_1 Z_4 = Z_2 Z_3$$

$$(R_1 + j\omega L_1) \frac{1}{j\omega C_4} = \left(R_2 + \frac{1}{j\omega C_2} \right) R_3$$

$$R_1 = \frac{C_4}{C_2} R_3$$

$$Q = \frac{\omega L_1}{R_1} = \omega R_2 C_2$$

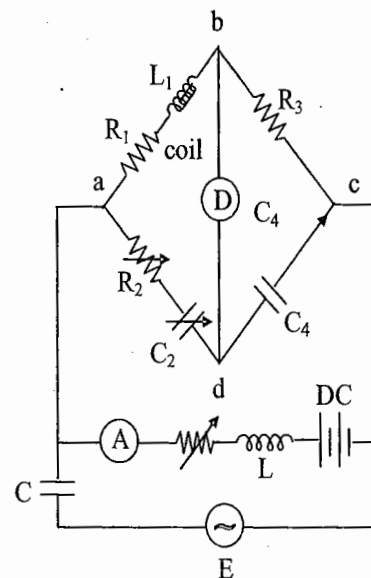
- This Bridge is suitable for medium Q-factor measurement.
- Owens bridge is used for incremental inductance measurement from known values of the fixed capacitance (C_4)



AIR CORE	IRON CORE
$L_{air} = \frac{N^2 \mu_0 A}{l}$	$L_{Iron} = \frac{\mu_0 \mu_r N^2 A}{l}$
	Physical Properties
	$L_{Iron} = L_{air} \times \mu_r$
	$L = \frac{N^2 \mu_0 A}{l} \times \mu_r$
	$\mu_r > 1$

$\mu_r \rightarrow$ Incremental Inductance

Modified Bridge for measurement of incremental Inductance



Note: The purpose of using the D.C. supply is to compensate the saturation effect of coil by supplying D.C. current

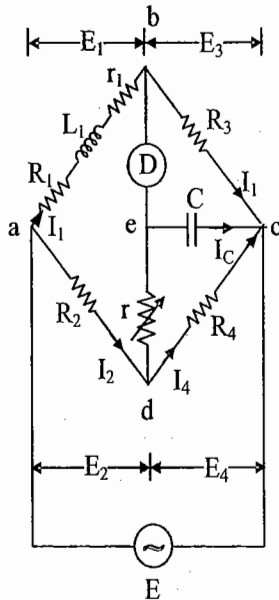
- Capacitor "C" blocks DC current. So It's used to block the DC current entering into AC source.



- Inductor "L" blocks AC. It is used to block AC current entering into the DC source.

ANDERSON'S BRIDGE:

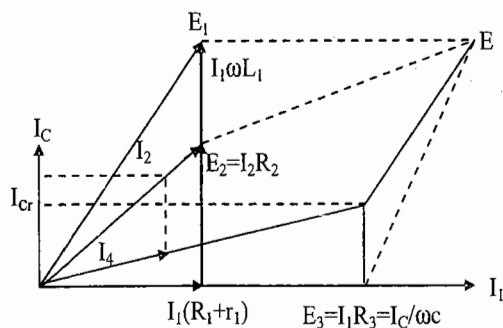
It is called as "5-point Bridge"



Under balanced condition

$$I_1(r + R_1 + j\omega L_1) = I_2 R_2 + I_C r \text{ and}$$

$$I_C \left(r + \frac{1}{j\omega C} \right) = (I_2 - I_C) R_4$$



$$I_1(r + R_1 + j\omega L_1) = I_2 R_2 + I_1 j\omega C R_3 r \text{ (or)}$$

$$I_1(r + R_1 + j\omega L_1 - j\omega C R_3 r) = I_2 R_2$$

$$j\omega C R_3 I_1 \left(r + \frac{1}{j\omega C} \right) = (I_2 - I_1 j\omega C R_3 r) R_4$$

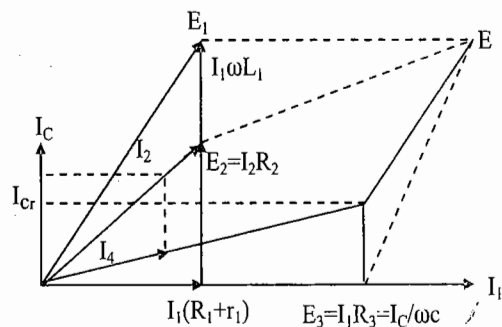
$$I_1(r + R_1 + j\omega L_1 - j\omega C R_3 r)$$

$$= I_1 \left(\frac{R_2 R_3}{R_4} + \frac{j\omega C R_2 R_3 r}{R_4} + j\omega C R_3 R_2 r \right)$$

$$R_1 = \frac{R_2 R_3}{R_4} - r_1$$

$$L_1 = C \frac{R_3}{R_4} (r(R_4 + R_2) + R_2 R_4)$$

- Anderson bridge uses a fixed capacitance 'C' and for the bridge balance, the resistances r & r1 are selected as variables.
- These are used for measurement of Low Q coil
- This bridge used for accurate measurement of Inductance over 'mH to wide range'
- It is the modification of Maxwell inductance capacitance bridge.
- Bridge will give two balance equations because of 'fixed point'



$$V_1 = V_2 + I_C r$$

$$V_{bc} = I_1 R_3 = V_3$$

$$V_{bc} = \frac{I_C}{j\omega C} = \frac{I_C}{\omega_c} \angle -90$$

$$V_{dc} = I_4 R_4 = V_{de} + V_{ec}$$

$$= I_C r + V_3$$

$$I_2 = I_C + I_4$$



Disadvantages

- (1) more complex bridge
- (2) shielding of the bridge is complicated

To measure the mutual – Inductance:

- (1) Heavy side campbell bridge
- (2) Carey foster bridges are used.

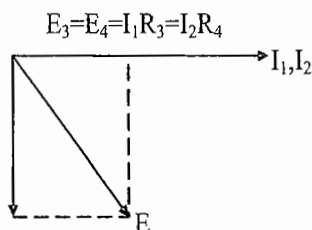
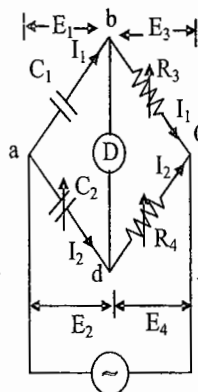
Capacitance Measurement

Capacitance can be measured from the following three bridges.

- (1) De-sauty's Bridge
- (2) Modified De-sauty's Bridge
- (3) Schering Bridge

De-sauty's Bridge

This bridge is suitable for perfect capacitor value measurement only (No-dielectric losses)

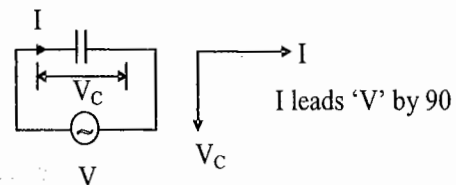


Under bridge balance condition

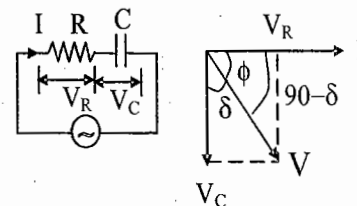
$$\frac{1}{j\omega C_1} R_4 = \frac{1}{j\omega C_2} R_3$$

$$C_1 = C_2 \frac{R_4}{R_3}$$

Pure capacitor (No-dielectric loss)



Practical capacitor (with dielectric loss)



$\sin \delta = \text{p.f of capacitor}$

$$\tan \delta = \frac{V_R}{V_C} = \frac{IR}{IX_C} = \omega RC$$

D-factor = $\tan \delta = \omega RC$

(or)

Derating factor (or) Dissipation factor

Example 2:

The four arms of a bridge are :

Arm ab: an imperfect capacitor C_1 with an equivalent series resistance of r_1

Arm bc: a non- inductive Resistance R_3

Arm cd; a non – inductive resistance R_4

Arm da; an imperfect capacitor C_2 with an equivalent series resistance of r_2 series with a Resistance R_2

A supply of 450Hz is given between terminals "a" and "c" and the detector is connected between



“b” and “d”. At balance: $R_2 = 4.8\Omega$; $R_3 = 2000\Omega$; $R_4 = 2850\Omega$ and $C_2 = 0.5\mu\text{F}$ and $r_2 = 0.4\Omega$ calculate the value of C_1 and r_1 and also find the dissipating factor for this capacitor.

(a) $0.712\mu\text{F}$; 3.65Ω ; 0.000734

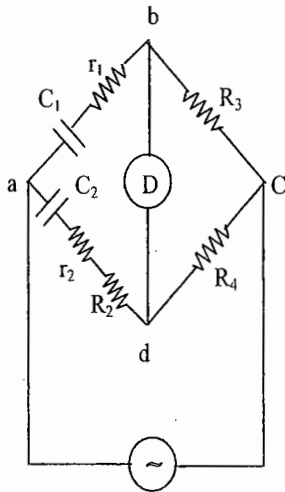
(b) $0.712\mu\text{F}$; 0.365Ω ; 0.00734

(c) $0.0712\mu\text{F}$; 3.65Ω ; 0.00734

(d) $0.712\mu\text{F}$; 3.65Ω ; 0.00734

Ans: (d)

Sol:



at balance

$$\left(r_1 + \frac{1}{j\omega c_1}\right)R_4 = \left(r_2 + R_2 + \frac{1}{j\omega c_2}\right)R_3$$

Separating the real and imaginary terms

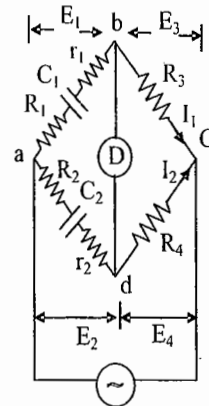
$$r_1 = (r_2 + R_2) \frac{R_3}{R_4} = (0.4 + 4.8) \times \frac{2000}{2850} = 3.65\Omega$$

And
$$C_1 = C_2 \frac{R_4}{R_3}$$

$$= (0.5 \times 10^{-6}) \times \frac{2850}{2000} \text{ F} = 0.712\mu\text{F}$$

$$\begin{aligned} \text{Dissipating factor } D_1 &= \tan\delta_1 = \omega C_1 r_1 \\ &= 2\pi \times 450 \times 0.712 \times 10^{-6} \times 3.65 \\ &= 0.00734 \end{aligned}$$

Modified De sauty's Bridge :



Used for measurement of C,D (dissipation factor)

$$C_1 = \frac{R_4}{R_3} C_2$$

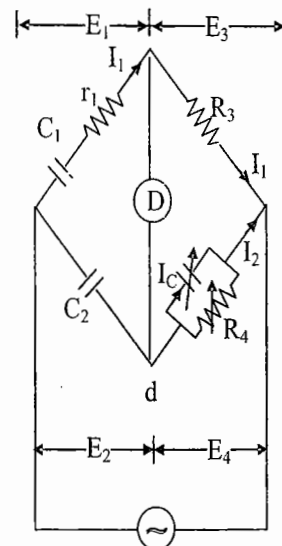
$$r_1 = \frac{R_3}{R_4} (R_2 + r_2) - R_1$$

Dissipation factor $\tan \delta = \omega C_1 r_1$

- this bridge is used for measurement of D factor practical capacitor

Schering Bridge:

This bridge is used for measurement of capacitance, D-factor (accurately), Dielectric loss, relative permittivity and properties of insulating materials.





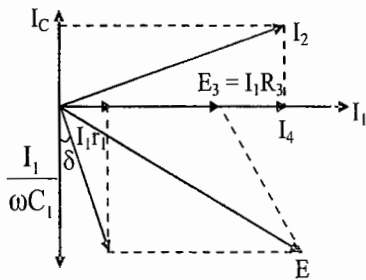
From the Bridge balance equations

$$C_1 = \frac{R_4}{R_3} C_2$$

$$r_1 = \frac{C_4}{C_2} R_3$$

$$D\text{-factor} = \omega C_1 r_1 = \omega \frac{R_4}{R_3} C_2 \cdot \frac{C_4}{C_2} R_3$$

$$D = \omega R_4 C_4$$



Example 3:

Common – data question Q(i) & Q(ii)

A dielectric sheet of thickness 1mm is tested at 50 Hz between two electrodes of 10 cm diameter. The Schering bridge employed has a standard compressed air capacitor C_2 of 100pF; a non – inductive resistor R_4 of 350Ω in parallel with a variable capacitor C_4 and a non – inductive variable resistor R_3 . At balance $C_4 = 0.4\mu\text{F}$, $R_3 = 250\Omega$.

(i) Find the power factor of the specimen

- (A) 0.04 (B) 0.18
(C) 0.23 (D) 0.12

Ans: (A)

Sol: Power factor $\sin \delta = \tan \delta = \omega C_4 R_4$

$$= 2\pi \times 50 \times 0.4 \times 10^{-6} \times 350$$

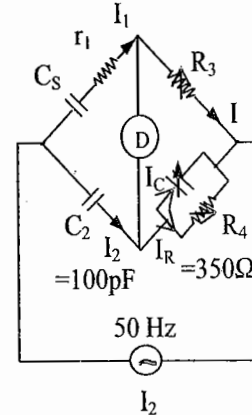
$$= 0.043$$

(ii) Find the permittivity of the sheet.

- (A) 1 (B) 2
(C) 3 (D) 4

Ans: (B)

Sol:



under balance

$$Z_s Z_4 = Z_2 Z_3$$

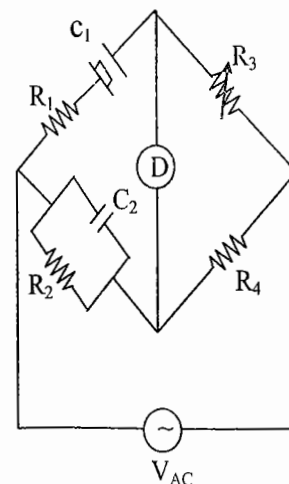
$$C_s = \frac{R_4}{R_3} C_2$$

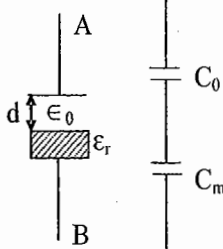
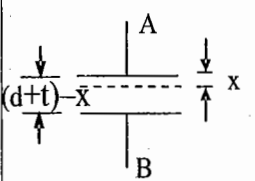
$$= \frac{350}{250} \times 100 \times 10^{-12} = 140 \times 10^{-12} \text{ F}$$

$$\text{Permittivity } \epsilon_r = \frac{C_s \times d}{\epsilon_0 \times A}$$

$$= \frac{140 \times 10^{-12} \times 0.001}{8.854 \times 10^{-12} \times \frac{\pi}{4} (0.1)^2} = 2$$

Measurement of (ϵ_r) relative permittivity:



 $d \rightarrow$ Distance between electrode specimen $t \rightarrow$ thickness of specimen $C_{AB} \rightarrow$ equalant capacitance	 Electrode is moved a distance of x (micrometer) to get same capacitance of equal to C_{AB}
$C_{AB} = \frac{C_1 C_2}{C_1 + C_2}$ $C_1 = \frac{\epsilon_0 A}{d}; C_2 = \frac{\epsilon_0 \epsilon_r A}{t}$ $C_{AB} = \frac{\epsilon_0 \epsilon_r A}{(t + d\epsilon_r)} \rightarrow (a)$	$C_{AB} = \frac{\epsilon_0 A}{(d+t)-x} \rightarrow (b)$

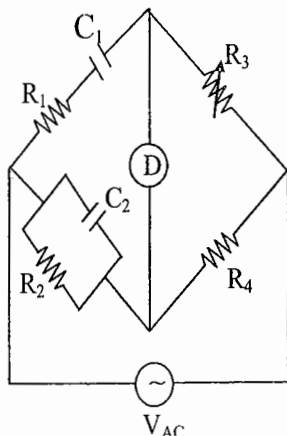
By equating equation "a & b"

$$\frac{\epsilon_0 \epsilon_r A}{(t + d\epsilon_r)} = \frac{\epsilon_0 A}{(d+t)-x}$$

$$\epsilon_r = \frac{t}{t-x}$$

Measurement of Frequency

Wien's Bridge:



from bridge balance

$$f = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \quad \frac{R_1}{R_2} + \frac{C_2}{C_1} = \frac{R_3}{R_4}$$

If $C_1 = C_2, R_1 = R_2$

$$\text{Then } \frac{R_3}{R_4} = 2; f = \frac{1}{2\pi R_1 C_1}$$

Applications:

- Used in distortion analyzer
- this bridge is not suitable for measurement of frequency of signals having harmonics, because bridge balance is difficult
- It's used for measurement of frequency from 100Hz to 100kHz

Causes of errors in AC bridges:

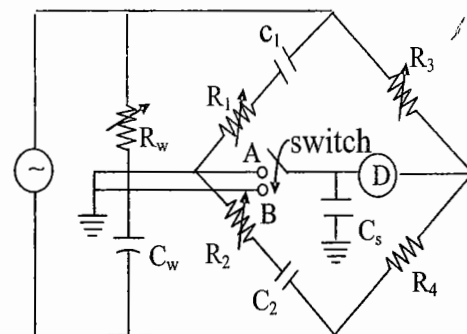
Stray conductance effect: Because of improper insulation leakage currents are present

Mutual inductance effect: The magnetic field coupling between the components

Stray capacitance effect: The capacitance forming between arms

Eddy current Error

Wagner Earth / Ground connections:





This connection is used to eliminate the stray capacitance produced between "Bridge arms & arms to the ground"

Procedure:

- Switch is kept on position A, make bridge balanced during this process C_s (stray capacitance) is charges to the certain potential.
- Now switch is moved to position 'B' by adjusting the Wagner resistance R_w . Bridge is balanced
- This indicates potential of C_s is equal to ground potential
- This process will continue either position of the switch i.e at A (or) B. the galvanometer shows zero deflection
- This corresponding to potential of C_s is equal to ground potential so that stray capacitance effect is reduced.

Resistance Measurements:

Resistance materials:

Manganian $\rightarrow$ Cu $\rightarrow$ 84% , Ni $\rightarrow$ 12%, mg $\rightarrow$ 4% and $\alpha = 0.00015/^\circ\text{C}$.

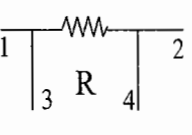
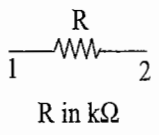
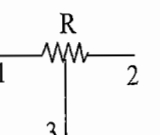
Constantan $\rightarrow$ Cu $\rightarrow$ 88%, Ni $\rightarrow$ 12% and $\alpha = 0.0017/^\circ\text{C}$.

Both have low value of ' α ' so it won't change with temperature

Cu $\alpha = 0.004/^\circ\text{C}$ $\rightarrow$ high value it's changes with temperature

Classification of resistance:

Low Resistance	Medium Resistance	High Resistance
Resistance $\leq 1\Omega$	Resistance $R(1\Omega \text{ to } 100k\Omega)$	Resistance $R > 100k\Omega$
Ex: Ammeter internal resistance, all	Ex: DC machine shunt field	Ex: cable insulation resistance, all

motors, generators and transformer's winding resistances, Earthing resistance, all semi-conductor devices when forward biased	winding resistance, all heater coil resistances, human body resistance, voltmeter resistance (100Ω)	motors, generators and transformers winding's insulation resistance, all semi conductor devices when reverse biased
To measure low resistance 4 terminal required	To measure medium resistance only two terminals are required	To measure high resistance 3 terminals are required
		
1 & 2 are current terminals, 3 & 4 voltage terminals	1 & 2 are terminals	Guard terminal (3) is used to eliminate leakage currents.
Methods to measure it 1. Potentiometer method 2. Kelvin double bridge	Methods to measure it 1. voltmeter, ammeter method 2. sub situation method 3. wheat stone bridge 4. ohm meter Method	Methods to measure it 1. loss of charge method 2. meggar 3. mega ohm bridge 4. direct deflection



Example 4:

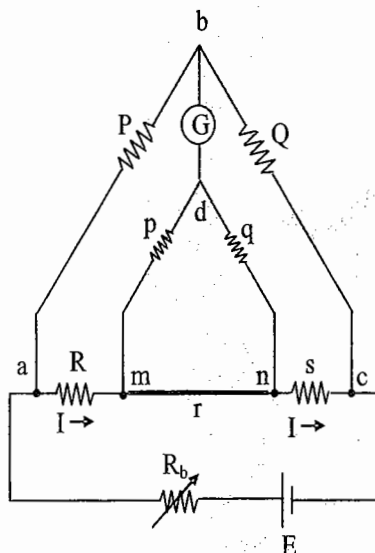
The methods for measurement of low and medium resistance cannot be used for measurement of high resistance as the

- (A) effect of thermo-electric e.m.f is considerable
- (B) effect of contact resistance is considerable
- (C) effect of connecting leads is considerable
- (D) resistance offered to the flow of current

Ans: (D)

Low Resistance Measurement

Kelvin's double bridge:



DC Bridge: source is DC so it is called DC Bridge

Under bridge balance conditions

$$R = \frac{P}{Q}(S) + \frac{qr}{p+q+r} \left(\frac{P}{Q} - \frac{p}{q} \right)$$

Including lead resistance

P, Q → External arm resistances

p, q → Internal arm resistances

s → standard variable resistance

r → terminals /contacts/leads resistance

R → unknown resistance

m → reversible switch to eliminate thermo electric emf

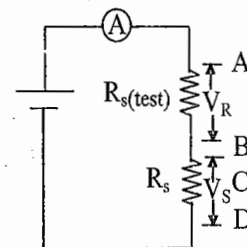
when lead resistance eliminated

$$R = \frac{P}{Q}S \quad \left(\because \frac{P}{Q} = \frac{p}{q} \right)$$

Advantage: It's used to eliminate the lead resistance effect so that suitable for measurement upto "micro ohms range"

Applications: Transformer winding.

Potentiometer method:



V_R = unknown resistance

V_S = known resistance

R_S = standard resistance

R_{Test} = test resistance

$$\therefore R = \frac{V_R}{V_S} R_S$$

Example 5:

Measurement of resistance by using potentiometer 4V is observed across the test resistance and 6V is observed across the standard resistance of 2mΩ. find test resistance?

Sol: $V_R = 4V$, $V_S = 6V$, $R_S = 2m\Omega$

$$R_T = \frac{V_R}{V_S} R_S \Rightarrow \frac{4}{6} \times 2m\Omega = \frac{4}{3} m\Omega$$



Note: difficulties in measurement of low resistance

- 1) terminals resistance
- 2) thermo electric emf's

Bifilar winding is used to reduce the inductance effect and aryton -perry winding is used to reduce capacitance effect.

Medium Resistance Measurement

V-A method	A-V method
$R_m = \frac{V}{A}$ V = Voltmeter reading A = Ammeter reading $= \frac{V_a + V_R}{I_R}$ $= \frac{I_R R + I_R R_a}{I_R}$ $R_m = R_a + R$ $R = R_m - R_a$ $\% \text{ error} = \frac{R_m - R}{R} \times 100$ $\% \text{ error} = \frac{R_a}{R} \times 100$	$R_m = \frac{V}{A}$ V = Voltmeter reading A = Ammeter reading $R_m = \frac{V}{I}$ By nodal $R = \frac{R_v R_m}{R_v + R_m}$ From above equation $\frac{R_m - R}{R} = \frac{-R_m}{R_v}$ Assume $R_m \approx R_T$ $\% \text{ Error} = \frac{R_m - R}{R} = \frac{-R_T}{R_v} \times 100$
The measured resistance is more than the actual value due to ammeter resistance	The measured resistance is less than the actual value due to voltmeter resistance
To reduce % error, V-A connection is	To reduce % error A-V connection is suitable

suitable for high resistance measurement in medium scale range

for low resistance measurement in medium scale range

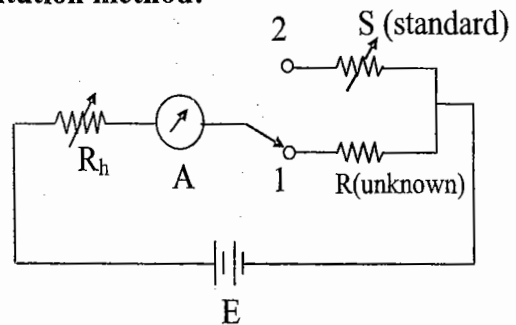
- To get same error on both V-A & A-V methods the value of unknown resistance "R" is equate both % errors

$$\frac{R_a}{R} = \frac{R}{R_v}$$

$$R = \sqrt{R_a R_v}$$

Note: in any method, the error due to only load side instrument

Substitution method:



Switch at position (1): vary the value of Rheostat 'R_h', such that the ammeter pointer reaches, the

chosen scale value. Then $I_1 = \frac{E}{R_h + R_a + R}$

Switch at position (2): Then current flowing through ammeter is I₂.

Equation $I_1 (R_h + R_a + R) = I_2 (R_h + R_a + S)$,

$$R = \frac{I_2}{I_1} [R_h + R_a + S] - (R_h + R_a)$$

If $I_1 = I_2$ (by $R_T = S$ ue)

If instead of ammeter, galvanometer is used then

$$R = \frac{\theta_2}{\theta_1} [R_h + R_a + S] - [R_h + R_a]$$



Example 6:

In a measurement of resistance by substitutions method a standard $0.5\text{M}\Omega$ resistor is used. The galvanometer has a resistance of $10\text{K}\Omega$ and gives deflection as follows

- (i) With standard resistor, 41 divisions
- (ii) With unknown Resistance 51 divisions

Find the unknown resistance?

- (a) $0.2\text{M}\Omega$ (b) $0.4\text{M}\Omega$
- (c) $0.5\text{M}\Omega$ (d) $0.8\text{M}\Omega$

Ans: (b)

Sol: let S = Standard Resistance

R = unknown Resistance

G = Galvanometer resistance

θ_1 = deflection with standard resistor

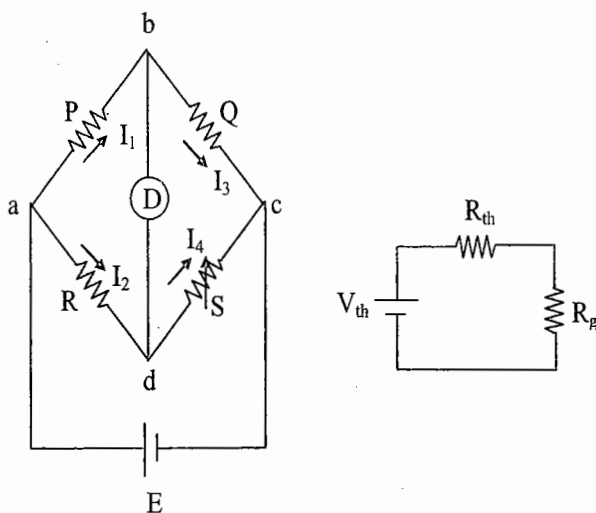
θ_2 = deflection with unknown resistor

$$\therefore \frac{\theta_1}{\theta_2} = \frac{R + G}{S + G}$$

Hence, unknown Resistance

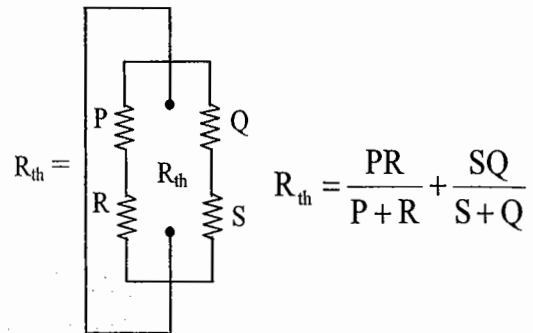
$$\begin{aligned} R &= (S + G) \frac{\theta_1}{\theta_2} - G \\ &= (0.5 \times 10^6 + 10 \times 10^3) \left[\frac{41}{51} \right] - 10 \times 10^3 \\ &= 0.4 \times 10^6 \Omega = 0.4\text{M}\Omega \end{aligned}$$

Wheat stone Bridge:



$$V_{th} = V_{bd} = V_b - V_d$$

$$= V \left[\frac{S}{S+Q} - \frac{R}{P+R} \right] = V \left[\frac{SP - RQ}{(S+Q)(P+R)} \right]$$



$$R_{th} = \frac{PR}{P+R} + \frac{SQ}{S+Q}$$

$$I_{th} = I_g = \frac{V_{th}}{R_{th} + R_g}$$

Under balanced condition

$$I_g = 0$$

$$SP - RQ = 0$$

$$SP = RQ$$

$$R = \frac{P}{Q} S$$

- Measured resistance 'R' is independent of galvanometer internal resistance and source internal resistance
- Hence the measured value is more accurate
- **Current sensitivity (S_i)** : It's the ratio of deflection of the galvanometer to that of the current flowing through the galvanometer

$$S_i = \frac{\theta}{I_g} \text{ mm}/\mu\text{A}$$

- **Voltage sensitivity:** It's ratio of deflection of galvanometer to that of the voltage across the galvanometer

$$S_v = \frac{\theta}{V_{th}} \text{ mm}/\text{V}$$



- **Bridge sensitivity:** $S_B = \frac{\theta}{\left(\frac{\Delta R}{R}\right)} \text{ mm}$

Note: In case of Bridge circuits sensitivity is an important parameter compared to accuracy resolution, precession etc

For maximum bridge sensitivity $S_B = \frac{V.S_V}{4}$

Example 7:

In wheat stone Bridge, with the added advantage of doubling the sensitivity of the bridge, which of the following effects on the measurement is eliminated

- (1) Resistance of connecting leads
- (2) Parasite emf's
- (3) temperature effect
- (4) Thermo electric emf

- (a) 1 and 4
- (b) 1 and 3
- (c) 2 and 4
- (d) All the above

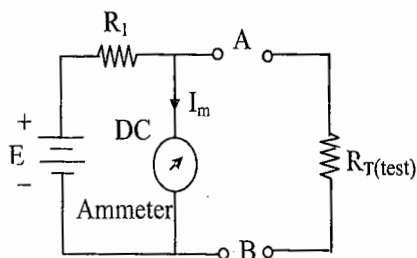
Ans: (c)

Ohm meter method:

Two types of methods

- (1) series ohm meter
- (2) shunt ohm meter

Shunt ohm meter:

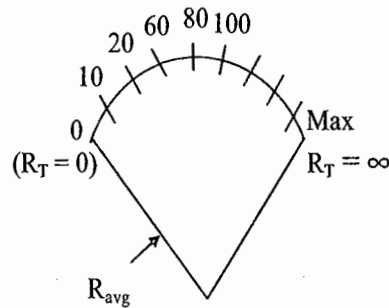


- (i) $R = \infty$ (O.C)

$$I_m \rightarrow \max$$

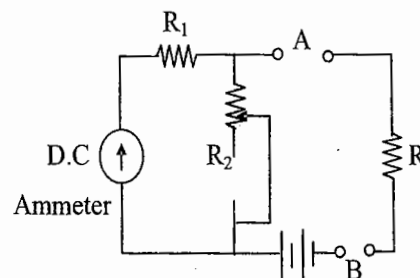
- (ii) $R = 0$ (S.C) $\Rightarrow I = 0$

$$I \propto R$$



- It is basically a **PMMC** instrument
- It works only for current
- The current scale is calibrated in terms of unknown resistance (R)
- It is used in continuity checking of electro static components

Series ohm meter:



$R_2 \rightarrow$ zero adjusting resistor

- (i) $R = \infty$ (O.C), $I = 0$

$$I_m \rightarrow \max$$

- (ii) $R = 0$ (S.C) $\Rightarrow I = \infty$

(full scale deflection)

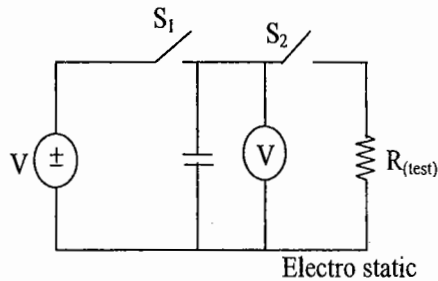
$$I \propto \frac{1}{R}$$

- In series ohm meter, the left extreme end is marked as ' $\infty\Omega$ ' and right extreme end is marked as ' 0Ω '.

Note: In series ohm meter, if battery is replaced with P.M.D.C generator then it's called as "meggar"

High resistance Measurement

(1) loss of charge method



'V' is initial voltage across capacitor

→ voltage across capacitor of any time

$$V_c = V \cdot e^{-t/RC}$$

$$R = \frac{0.4343t}{C \log_{10}(V/V_c)}$$

t = seconds

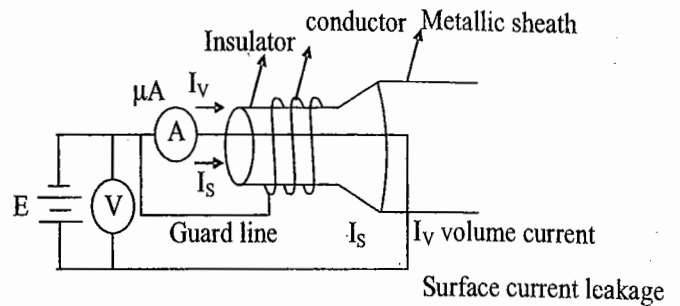
- It is used to find cable insulation resistance in researching laboratories

Example 8:

In loss of charge method for measuring insulation resistance of cable, voltage across capacitance of $0.4343 \mu\text{F}$ is dropped from 200V to 20V in (one minute). Find the insulation resistance value?

$$\text{Sol: } R = \frac{0.4343 \times 1 \times 60}{0.4343 \times 10^{-6} \log_{10}\left(\frac{200}{20}\right)} = 60 \text{ M}\Omega$$

Direct –deflection method



I_v → volume current, passing through the conductor through the insulation and through metallic sheath

I_s → surface leakage current passing through the conductor, over the insulation under the metallic sheath.

Ammeter is reading both I_v & I_s . hence the measured resistance has an error.

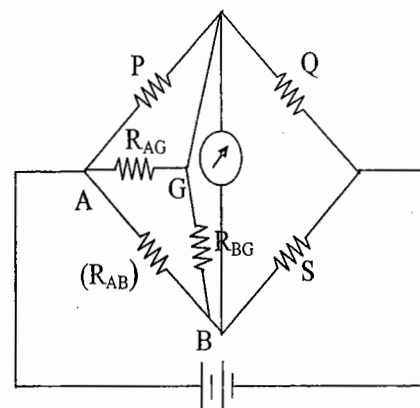
$$I = I_v + I_s$$

Guard wire: It's a conductor with small number of turns, tightly connected over the insulation and maintained at supply terminals.

$$R = \frac{\text{voltmeter reading}}{\text{Ammeter reading}}$$

Ammeter reading is only I_v

Mega Ohm Bridge:



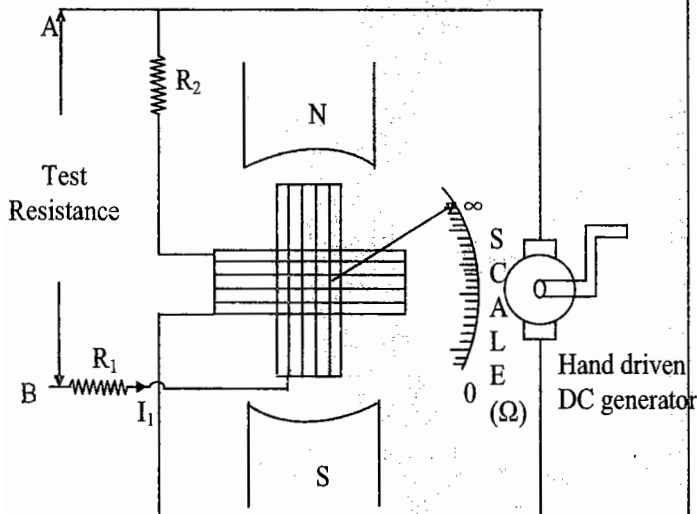


$$R_{AG} || P = P_1$$

- R_{BG} Galvanometer resistance $\rightarrow$ resultant resistance is very less then only R_{AB} is present
- Then $R_{AB} = \frac{P_1}{Q} S$
- Guard terminal of source to eliminate resistances produced by the leakage currents inside the cable insulation



Electro dynamo-meter meggar



- Potential coil (P.C) and current coil (C.C) are moving and these coils are connected such that their corresponding torques acts in opposite direction i.e both P.C and C.C are rigidly connected 90° .
- Resistors R_1 & R_2 are connected in series with P.C & C.C
- Here left extreme end ' ∞ ' is marked on a scale and '0' is marked on right extreme end of scale.

P.C

$$F_1 = BI_1 l \sin \theta$$

C.C

$$F_1 = BI_1 l \sin(90 + \theta)$$

$$= BI_1 l \cos \theta$$

When pointer gives reading

$$F_1 = F_2$$

$$BI_2 l \sin \theta = BI_1 l \cos \theta$$

$$\tan \theta = \frac{R_2}{R_1 + R}$$

$$\theta \propto \frac{1}{R}$$

$$\theta = 0 ; R = \infty$$

$$\theta = \infty ; R = 0$$

- The meggar voltage ratings are 500V, 1000V, 2 kV, 2-5kV,
- The minimum application voltage is 100V

Applications:

- Used for measurement of Insulation resistance of cables
- Winding insulation resistance of motors, generator, Transformers bushings

Note:

- In meggar spring control is not required

Example 9:

The controlling torque in a meggar is provided by

- (a) springs
- (b) Weights attached to the moving system
- (c) It does not need any controlling torque
- (d) None of the above

Ans: (c)